Dianchao Wu

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Education

Iowa State University

B.S. Mathematics — Expected May 2027

Cumulative GPA: 3.85/4.00 · **Mathematics GPA:** 3.89/4.00

Research Experience

Degenerate Diffusions in Finite and Infinite Dimensions: Smoothing and Convergence

Student Researcher · Advisor: David P. Herzog — Summer 2026-Present

NSF-funded research · Award DMS-2246491

- Formulated the Berry-Esseen framework for the unwrapped displacement and proved the overdamped results.
- Proved the main underdamped theorem and supporting propositions via a direct Markov-additive treatment of the displacement; verification of Assumption (A2) was carried out by O. Smith.
- Proved in Proposition 11.1 that, for one-dimensional periodic underdamped dynamics started from equilibrium and sufficiently small positive γ, the Berry-Esseen constant C_γ can be chosen with $C_\gamma \le C\gamma^{-1/2}$, where C is independent of γ.
- Simulated overdamped and underdamped transport, measured the friction-dependent constants, and compared them with the averaged energy dynamics.

The joint manuscript also includes additional numerical analysis by A. Valse.

Random Prepayment Sensitivity in Nonlinear MBS Pricing

Student Researcher · Advisor: Rana D. Parshad — 2026-Present

- Analyze bounded continuous prepayment perturbations in the Papi-Briani MBS model at $p = 2$ using viscosity theory and a Lipschitz BSDE representation.
- Established a bounded linear price response with a uniform second-order remainder, yielding Fréchet differentiability and mean-square sensitivity.
- Derived centered-noise mean-bias and Hilbert-space variance expansions; linked the leading price variance to the input-noise covariance.
- Planned for Spring 2027: extend the MBS evolution to an SPDE with additive noise; study well-posedness, moment bounds, stability, small-noise limits, and numerical simulation.

Random Matrix Theory and Quantum Expanders

ISmART Research Project · Mentor: Dr. Jananan Arulseelan — Spring-Summer 2026

Past exploratory study of Haar-distributed unitary matrices and quantum expanders.

Manuscripts

Berry-Esseen Bounds for Periodic Langevin Dynamics

Coming soon

O. Smith, A. Valse, D. Wu (alphabetical). Mentor: Professor David P. Herzog.

Random Prepayment Sensitivity in Nonlinear MBS Pricing

Working manuscript

Directed Studies

Four faculty-supervised reading courses.

Stochastic Calculus | David P. Herzog

Text: Bernt Øksendal, *Stochastic Differential Equations: An Introduction with Applications*, 6th ed.

Brownian motion, Itô integration and formula, SDEs, diffusion generators, and stochastic dynamics.

Foundations of Probability | Ruoyu Wu

Text: Ross and Leadbetter, *A Basic Course in Measure and Probability*.

Measure-theoretic probability, random variables, integration, independence, and modes of convergence.

Real Analysis and Measure Theory | Ruoyu Wu

Text: H. L. Royden and P. M. Fitzpatrick, *Real Analysis*, 4th ed.

Lebesgue measure and integration, convergence theorems, and L^p spaces.

Partial Differential Equations I | Rana D. Parshad

Text: Lawrence C. Evans, *Partial Differential Equations*.

Sobolev spaces, weak derivatives and solutions, and functional-analytic methods for PDE.

Selected Coursework

In progress · Fall 2026

MATH 5010 - Introduction to Real Analysis; **MATH 5540** - Stochastic Process Models.

Planned · Spring 2027

MATH/STAT 6450 - Advanced Stochastic Processes; **MATH 5250** - Numerical Analysis of High-Performance Computing; **STAT 6900A** - Advanced Special Topics: Continuous Time Financial Modelling and Its Estimation.

Selected coursework · Completed

MATH 3850 · Introduction to Partial Differential Equations; MATH 4030 · Intermediate Abstract Algebra; MATH 4430X · Introduction to Mathematical Finance; MATH 3010 · Abstract Algebra I; MATH 3170 · Theory of Linear Algebra; MATH 4900 · Independent Study, 6 credits.

Honors & Awards

GCAA Cobalt Golf All-America Scholar (NCAA Division I)	2025-26
CSC Academic All-District	2025, 2026
Fred Wright Mathematics Endowed Scholarship	May 2026
Academic All-Big 12 First Team	2025, 2026
ISU Athletics O.S.C.A.R.S. Award	May 2026
ISU Student-Athlete of the Month	September 2026
Academic All-Big 12 Rookie Team	2024
Big 12 Commissioner's Honor Roll	Fall 2023-Spring 2026

Athletics

Iowa State University Men's Golf

NCAA Division I Student-Athlete — August 2023-Present

Represent Iowa State University in NCAA Division I tournaments. Balance more than 20 hours of weekly training and competition with a full mathematics course load. Best World Amateur Golf Ranking: 443.

Technical Skills

Programming: Python, MATLAB, C (currently learning)

Scientific work: Numerical simulation, stochastic-process simulation, market-data analysis

Typesetting: LaTeX

Languages: English (fluent); Mandarin and Hakka (native)